



MATH 251 CALCULUS III

SYLLABUS – Summer, 2026

Professor: Dr. Frank Forte

Office: West 320

E-mail: Frank.Forte@raritanval.edu

CONTACT INFO: The best way to contact me is to use the Canvas email tool. That will allow me to respond to you in the most efficient way. Note that I cannot send messages to personal (non-RV) email addresses. Allow 24 hours (weekdays) for a response.

Office Hours: As announced

IMPORTANT NOTE: Detailed information about RVCC Student Resources and College Policies is available in **Canvas** under the link [RVCC Student Resources](#). This includes information about technology support, counseling, accommodations and/or services to persons with disabilities, academic integrity, and more. You can also access it by going to this website: <https://rvcc.instructure.com/courses/3795531>.

Math & Computer Science Department Chair: Prof. Aditi Patel (Aditi.patel@raritanval.edu)
Room W318

CLASS TIME: Class meet online asynchronously. There are no in-person meetings on campus.

Textbook: *Calculus Early Transcendentals, 3rd edition*, by Briggs & Cochran, Publishers: Pearson Addison Wesley. NOTE: You have already paid for online access to your textbook in your tuition. In Canvas, go to the first module and find the link "Pearson Course Materials." That is how you will set up your access to MyMath Lab and the online HW. This will give you access to your online HW, textbook, and numerous resources including a Student Solution Manual for practice problems in the textbook.

Technology:

- MyMathLab (integrated into Canvas): for all online graded homework.
- Software for labs: Mathematica and Desmos available free online.
- Graphing calculator, TI-84/TI-84 plus is optional for labs and homework. Graphing calculators are not allowed on tests and the final exam.
- Scientific calculator for assignments and tests.

The MML course will be linked with Canvas. Thus, there is no MML course ID code. Please don't try to register by searching for the course on MyMathLab. If you don't link Canvas to MML, none of your homework grades will transfer!

Student Learning Outcomes

The learning outcomes for all math courses are linked to the NACE competencies for career readiness. More information about their career competencies can be found here:

<https://naceweb.org/career-readiness/competencies/career-readiness-defined>

The table below identifies both your course learning outcomes for Calculus III as well as their corresponding career competency:

Course Learning Outcomes	NACE Competency
1. Compute with vectors in three-dimensional space	Critical Thinking Technology Teamwork
2. Differentiate functions of several variables, including graphing functions of two variables.	Critical Thinking Technology Teamwork
3. Evaluate multiple integrals.	Critical Thinking Technology Teamwork
4. Utilize Green's Theorem to evaluate line integrals.	Critical Thinking Technology Teamwork

AI Use Statement for this Course [Instructors may edit. This may also be superseded by new college policies that may be developed. If you are using AI in the teaching of the course, you should state where and how it will be used.]

Artificial Intelligence (AI) tools, including but not limited to ChatGPT, Wolfram Alpha, and other computer algebra systems, can be powerful aids in learning and exploring upper level math courses. However, their use must align with the learning objectives and academic integrity policies of this course.

Permitted Uses:

- Using AI to review concepts, definitions, and theorems.
- Getting help with understanding problem-solving steps after attempting a problem.
- Checking your work or exploring multiple approaches **after** completing your own solution.
- Using AI as a tutor to reinforce understanding of difficult topics.

Prohibited Uses:

- Submitting AI-generated solutions as your own without doing the work yourself.
- Using AI to complete quizzes, tests, or any assessments unless explicitly allowed.
- Copying AI-generated responses without understanding the underlying concepts.

Remember: The goal of this course is for you to develop problem-solving skills and mathematical understanding. Overreliance on AI can hinder your progress and may result in academic dishonesty consequences.

Attendance, Withdrawal and Attendance Verification:

All students are expected to log in daily. See the policies on Canvas for more information. The last day to withdraw this semester is August 5.

If you do not attend class (or your instructor is unable to verify your participation/attendance) prior to the attendance verification date of **July 9**, you will be DROPPED from the course. Logging on to Canvas does not count as attendance. You will need to complete at least one assignment / discussion to avoid being dropped.

Canvas: Canvas is the Learning Management System for RV. Its a place where you will find course documents, assignments, check grades and complete your online homework assignments. There is no My Math Lab course ID – its integrated into Canvas.

GRADING POLICY:

Your grade for the course will be calculated by considering:

Tests (3)	40%
Homework (online)	10%
Discussion Posts	7%
Paper Labs	13%
Cumulative Final Exam	30%

Your percentage grade will be converted to a letter grading using the following scale:

A [90, 100]	B+ [86, 89]	B [80, 85]	C+ [76, 79]
C [70, 75]	D [60, 69]	F [0, 59]	

Note: While a D will enable you to graduate, a grade of C or better is required for transfer.

- **HOMEWORK:** Practicing and completing problems are essential to learning the material in this course. Work will be completed online using the My Math Lab program. After the due date, I will transfer those grades over to Canvas so that you can keep tabs on your grades using Canvas. These problems will take time and effort to complete, but they are important to your learning process. I suggest that you keep a notebook where you show your work so you can refer back to it when you prepare for the test.

The due dates for the MML assignments are listed on the syllabus and will close at 11:59:00pm that day. At that time they will be closed and will not be reopened without an acceptable reason (severe illness, accident, childbirth, alien abduction, etc.).

- **LABS:** This class has time each week designated for labs. You are encouraged to get help from me on these labs and to go over your work with me before submitting them. Some are designed to be completed with the use of technology. Labs may be done in pairs, with one lab handed in for both participants. Labs must be organized and legible. You will scan your work and upload it to Canvas in a single Word or pdf file by the due date. Late work will not be accepted without documentation.

- **DISCUSSIONS:** There are graded discussions throughout the semester on Canvas. You should give a complete response to the questions posted and are encouraged to comment respectfully on your classmates' posts. The discussions are set up so that you will not see responses until you post your response, so if the discussion looks blank, it might just be waiting for your response. Also, please be aware of the college's policy on Academic Integrity when you post. In short, do NOT ask Chat GPT to make you a response to a discussion post. Using AI to look up facts is good. Asking it to write the whole response and then submitting it as your own is not.
- **TESTS: Tests are taken online using the Respondus Monitor program.** Each test has two dates, listed on the syllabus, For these online tests, you will need **a quiet space to take the test, a webcam, the ability to print pages, and the ability to scan and upload pages of work.** Before each test, you MUST print the work pages that are posted. You will use those pages for all graded work. See the video on Canvas about online testing for more information. Work must be uploaded within 15-20 minutes of submitting the test.

Note that the test proctoring system records both video and audio of the testing room and also records your computer screen. Make sure your face is visible in the camera. I will review the video, especially if inconsistencies are noticed. There are obviously ways to evade the system, but please note that if I suspect that irregularities have occurred, I will ask you to solve similar problems while I watch, either in my office or on Zoom and will expect similar results. For example, if you calculate the value of a triple integral with no work, then I will ask you to do the same while I watch. Instead of taking the time to figure out ways around the system, use that time to learn Calculus. Its worth it in many ways!

- Because you do not have to travel to campus, and the test hours are flexible, missing a test would be extremely rare. Thus, if you miss a test, I would expect that you will be able to provide a note from a doctor indicating that you were completely incapacitated during the days of the test (or perhaps abducted by aliens). Once the test has closed (and especially after they have been returned), a makeup is not likely even with a good reason. In general, late assignments (including discussions) will not be accepted.
- Only NON-GRAPHING calculators allowed for the tests. Note that it is important to show your work as you solve problems so that you can receive partial credit. The final exam will be cumulative and will contain material from all chapters covered.
- **FINAL EXAM:** The final exam is cumulative and will be taken online. Dates are listed on the syllabus.

COURSE MANAGEMENT

This is a fast paced, six week course that covers everything that a standard Calculus III course would cover in fifteen weeks. The college policy is that for every hour you are in class, you should spend two hours outside of class working on homework, completing labs, etc. If we were on campus, this course would meet about 12 hours a week in person, 24 hours outside of class. Therefore, **expect to spend about 36 hours per week covering the material, watching my videos, completing assignments, preparing for tests, etc.** It is your responsibility to keep up with the class work and homework assignments.

Pay careful attention to due dates. There is something due almost every day. The purpose is to keep you on track, to avoid the crush of working coming due all at once, and to give you regular feedback on your work so you can track your grades.

VISITING STUDENTS: In order to transfer credit back to your home four year institution, you will need to request RVCC to send a transcript to the school at the conclusion of your last summer class. Students should allow approximately five business days for a request to process.

STUDENTS WITH DOCUMENTED LEARNING DISABILITIES:

Raritan Valley Community College offers reasonable accommodation and/or services to persons with disabilities. Any student who has a documented disability and wishes to self-identify should contact the Center for Accessibility and Inclusive Education (CAIE) at (908) 526-1200 ext. 8534, or email caie@raritanval.edu. Accommodations are *individualized* and in accordance with Section 504 of the Rehabilitation Act and the Americans with Disabilities Act. In order to receive accommodations, students must be registered with the Center for Accessibility and Inclusive Education. Students should register with the office as soon as possible. No accommodation is official until the Accommodation Letter is issued from the student to their instructor. For more information, please visit the [CAIE website](#).

Final Things:

- As per the college guidelines, you should expect to spend two hours outside of class for every hour spent in class. This could be studying, working on homework, etc. Please see me if you are having trouble keeping up in the class.
- Please be aware of the college's updated **policies on Academic Integrity**. This applies not only to taking tests, but situations like the use of AI to complete labs and homework. If you have questions regarding appropriate use of technology, please ask.

MATH 251: CALCULUS III

Summer 2, 2026 SCHEDULE

This is a tentative schedule of the lecture topics, discussions, homework assignments and test dates. Any changes will be posted on Canvas.

DATE	SECTION & TOPIC	DUE (11:59 pm):
WEEK 1		
M 7/6	Course Introduction 13.1: Vectors in the Plane 13.2: Vectors in Three Dimensions	
T 7/7	13.3: Dot Product 13.4: Cross Product Lab 0: Intro. to Mathematica (not graded)	Discussion 1 (Introductions) Lab 0: Intro to Mathematica (not collected)
W 7/8	13.5: Lines and Planes in Space 13.6: Cylinders and Quadric Surfaces Lab 1: Sections 13.1 – 13.5	HW 13.1 – 13.3 (online MML) Discussion 2 (Vectors & Dot Prod.)
TH 7/9	14.1: Vector Valued Functions 14.2: Calculus of Vector Valued Functions	Lab 1 (Ch. 13)
SU 7/12		HW 13.4 – 13.6 Practice Quiz due
WEEK 2		
M 7/13	14.3: Motion in Space 14.4: Length of Curves Lab 2: Chapter 14	Discussion 3 (Potato Chips and 3D Surface)
T 7/14	14.5: Curvature and Normal Vectors Review sheet / practice problems	HW 14.1 – 14.2 Lab 2 due
W 7/15	15.1: Graphs and Level Curves 15.2: Limits and Continuity Work on review problems	
TH 7/16	TEST # 1: Chapter 13 & 14 – take online TH 7/16 or F 7/17	HW 14.3 – 14.5 Test 1 due by 10pm F 7/17
SU 7/19		HW 15.1 – 15.2
WEEK 3		
M 7/20	15.3 Partial Derivatives 15.4 The Chain Rule Lab 3 Sections 15.1-15.3	Discussion 4 (Partial Derivatives)
T 7/21	15.5: Directional Derivatives & Gradient Finish Lab 3 – due at night.	HW 15.3 – 15.4 Lab 3 due
W 7/22	15.6: Tangent Planes & Linear Approximation Start Lab 4: Sec. 15.3 – 15.5	
TH 7/23	15.7: Maximum & Minimum Problems Finish Lab 4 / Start Lab 5 (Sections 15.7 – 15.8)	Lab 4 due
SU 7/26		Discussion 5 (Hessian Matrix) HW 15.5 – 15.6

WEEK 4		
M 7/27	15.8: Lagrange Multipliers Finish Lab 5: Sections 15.6 – 15.8 Review for Test # 2 (rev. sheet)	Lab 5 due HW 15.7 – 15.8
T 7/28	TEST # 2: Chapter 15 – take T 7/28 or W 7/29	
W 7/29	16.1: Double Integrals over Rectangular Regions 16.2: Double Integrals over General Regions 16.3: Double Integrals over Polar Coordinates Lab 6: Start Lab 6 (Sections 16.1 – 16.4)	TEST 2 due by 10pm
TH 7/30	16.4: Triple Integrals 16.5: Triple Integrals in Cylindrical and Spherical Coordinates Finish Lab 6 – due Sunday night.	Discussion 6 (Double Integrals) HW 16.1 – 16.2
SU 8/2		Lab 6 due (16.1 – 16.3)
WEEK 5		
M 8/3	16.6: Integrals for Mass Calculations Start Lab 7 (Sections 16.5 – 16.7)	HW 16.3 – 16.5
T 8/4	16.7: Change of Variables in Multiple Integrals Finish Lab 7 – due Wed. night Review for Test # 3	Discussion 7 (Spherical, Jacobian)
W 8/5	TEST # 3: Chapter 16 – take W 8/5 or TH 8/6	Lab 7 due HW 16.6 – 16.7
TH 8/6	17.1: Vector Fields 17.2: Line integrals 17.3: Conservative Vector Fields	TEST 3 due by 6pm
SU 8/9		HW 17.1 – 17.2
WEEK 6		
M 8/10	17.4: Green's Theorem 17.5: Divergence and Curl Lab 8: Sections 17.1 – 17.3	Discussion 8 (VF & Line Integrals)
T 8/11	17.6: Surface Integrals 17.7: Stokes Theorem 17.8: Divergence Theorem Finish Lab 8	HW 17.3 – 17.4 Lab 8
W 8/12	Review Packet for Final Exam FINAL EXAM	HW 17.5 – 17.8
TH 8/13	FINAL EXAM – take Wed. 8/12 or Thurs. 8/13 Final exam is cumulative: chapters 13 to 17.	Final Exam due by 10pm on Thursday 8/13.